05 The Rate Constant

We saw in the last lecture that the rate law is equal to a rate constant k multiplied by some function of concentration. For example: $-r_A = k \cdot f(C_A, C_B, C_C, \ldots)$. In this lecture, we'll concern ourselves with the rate constant k.

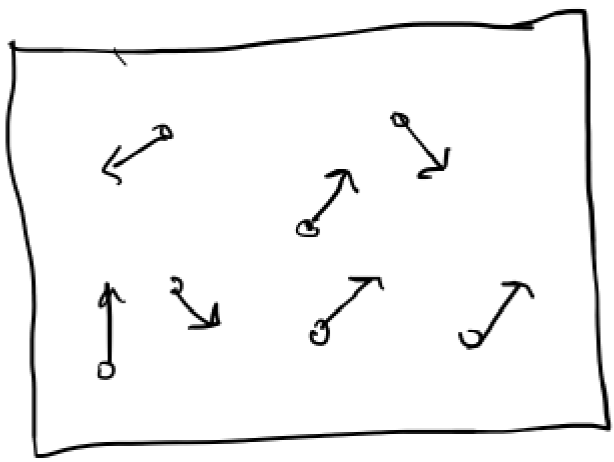
The rate constant follows an Arrhenius equation: $k_A(T) = A(T)\exp(-\frac{E}{RT})$, where $A(T)$ is the pre-exponential term or "frequency factor" and E is the activation energy.

$A(T)$ quantifies the frequency of occurrences that <u>could</u> lead to a chemical reaction

$\exp(-\frac{E}{RT})$ is the probability that such an occurrence indeed will result in a chemical reaction.

Specifically, $\exp(-\frac{E}{RT})$ is a Boltzmann probability, which is derived from statistical mechanics.

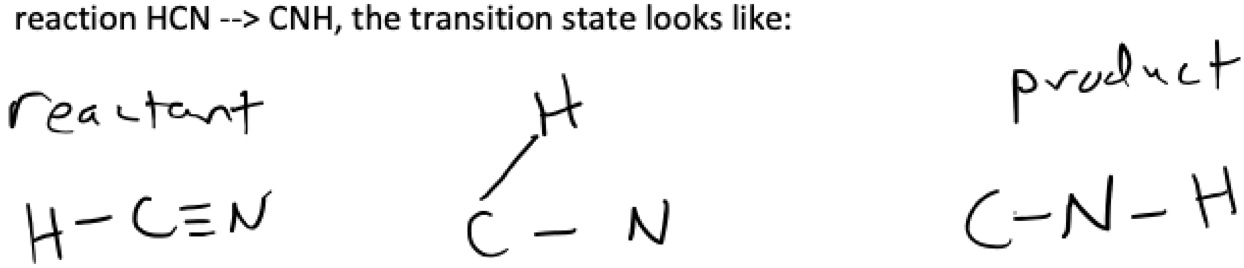
Example: In the gas phase, reactions happen when gas molecules collide. However, not every collision results in a reaction; only collisions with sufficient energy to surpass the activation energy result in a reaction. $A(T)$ quantifies the total the frequency of collisions, i.e., the number of collisions per unit time, and even accounts for those collisions that do not have sufficient energy to surpass the activation energy. The exponential term $\exp(-\frac{E}{RT})$ scales $A(T)$ by that fraction of collisions that have sufficient energy to surpass the activation energy.



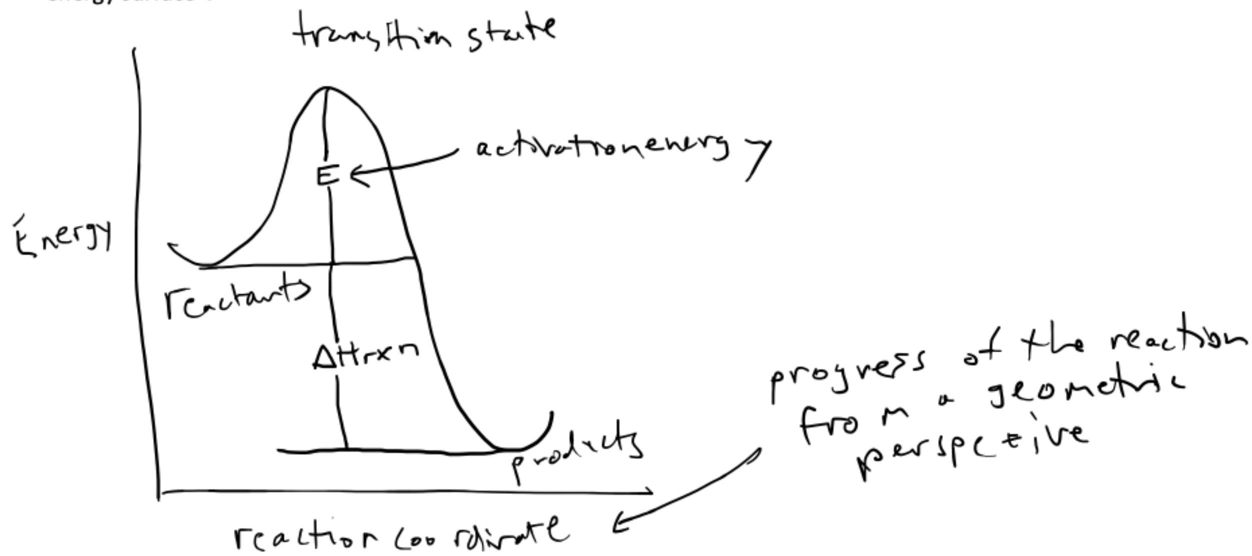
Gas molecules zoom around really fast, sometimes they collide. The frequency of collisions can be predicted using theorems from physical chemistry.

E is the activation energy. It is equal to the difference in enthalpy between the transition state and the reactants.

The transition state is a compound with chemical structure "in between" that of the reactants and products. For example, for the isomerization reaction HCN --> CNH, the transition state looks like:



Notice how the transition state is "contorted" compared to the reactants and products. This contortion costs energy … this energy cost gives rise to the activation energy. We can illustrate these energies on a "potential energy surface":



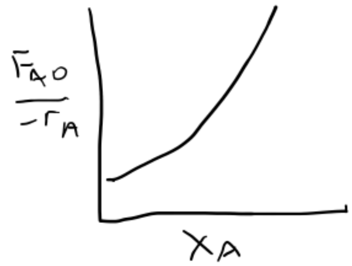
There is a lot more information on Canvas about transition states, if you are interested.

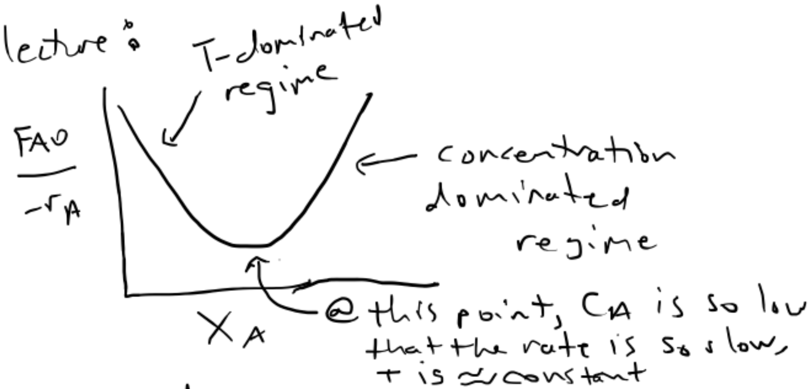
<u>Key point:</u> The rate expression depends on concentration and temperature. The temperature dependence comes from the rate constant *k*.

- $\exp(-\frac{E}{RT})$ increases exponentially with *T*. *E* is almost always ≥ 0. *E* is almost always constant with *T*.
- *A*(*T*) is often linear with *T* … Sometimes it is proportional to *T* to some small power, for example $T^{0.5}$ or T^2. In this class, we assume that *A* is constant with *T*.

<u>Question from the lecture on Levenspiel plots:</u> Why might $\frac{F_{A0}}{-r_A}$ decrease with X_A? Consider the case where $r_A = -kC_A^n$. $n > 0$

recall from Levenspiel lecture:





isothermal – no T dependence.

$\uparrow X_A \downarrow C_A \downarrow -r_A$

$\uparrow \frac{1}{kC_A^n}$

adiabatic – rate depends on T

consider an exothermic reaction

$\uparrow X_A \uparrow T \uparrow kC_A^n \downarrow \frac{1}{kC_A^n}$